

Postgraduate Application Form



Yerbury-Hodgson, Mr Oliver

Course MPhil in Machine Learning and Machine Intelligence
Department Department of Engineering
Course start date 01 Oct 2021 (MT 2021)

Date submitted
Mode of study Full Time

Academic History

Sep 2014 - Jul 2017 Bachelor of Science in Discrete Mathematics (Hons) (Computer Science) First (Hons) The University of Warwick (*United Kingdom*)

Immigration

Nationality United Kingdom (1st)
Country of birth United Kingdom
Currently ordinarily resident United Kingdom
Visa Not required

Language

Not required

Scholarships

Apply for funding Yes
Apply for Cambridge Trust Yes
Apply for Gates Cambridge No

Curriculum Vitae

oliver_cv_cambridge_2020.pdf

Career Goals

872/1000 chars

I'd like to apply Machine Learning and Machine Intelligence to topics on sustainability. Recently the area of agriculture technology, and the use of computer vision and sensors to monitor crops and increase yield, has demonstrated to me how viable technology can be in achieving the goal of reducing our global impact and addressing food shortages or waste. In addition to limiting the use of the Earth's resources, we must manage the growing population. Predicting COVID outbreak areas using web-scraping and Natural Language Processing demonstrates another way in which technological advancements are improving our well-being and safety while reducing our impact on the Earth.

I'd like to complement my experience in industry so far with the theoretical knowledge this course would provide. I believe this would equip me to play a leading technical role in this area.

Additional Information to Support Application

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Course Specific Questions

MLMI - MPhil track Computer Vision and Robotics; Speech and Language Technology

Core - Motivation for applying. In the modern world, the set of challenges are vast and complex, and the adoption of technological solutions that aim to address these challenges are often met with concerns from the public about data privacy, and issues with scalability. I believe that an education at Cambridge helps foster the analytical rigour required to produce not only good insight, but also to be able to communicate this effectively and convince others of the value of one's work. What has motivated me to apply to this course in particular, is that it is offered jointly with many different groups across Engineering, Computer Science, and Statistics, providing a holistic view on Machine Learning. In addition to this, modules such as Advanced Machine Learning teach others to read, reproduce, and contribute to research, skills that are very important given the pace that this field is moving. Finally, the course's core modules appear to explore many of the real-world applications that inspired me to apply in the first place. Computer vision always gains my interest in particular because of the augmentation of traditionally human tasks with machines.

Core - Brief statement of relevant work experience I have created a machine learning platform for my colleagues to perform their daily investigations, combining user interfaces, big data and Python libraries, and GPU compute capabilities into a single application. I take pride in the fact that I was able to understand the high-level architecture such a platform would require as well as the low-level implementation. Challenges such as why the GPU module hadn't loaded into the kernel, a question that drove me to delve into the OS level operations. Additionally, I have contributed to solutions with sentiment analysis and time-series modelling components. These roles were fulfilled using distributed computing solutions such as Hadoop and Apache Spark, working jointly with data scientists.

MLMI - Academic preparation (math) MA124 - Mathematics by Computer (Differentiation, PDEs) MA137 - Mathematical Analysis (Differentiation) CS137 - Discrete Mathematics and its Applications II (Fourier Series)

MLMI - Academic preparation (algebra) MA106 - Linear Algebra (vectors, matrices, linear transformations, matrix inversion) MA251 - Algebra I: Advanced Linear Algebra (eigenvalues and eigenvectors)

MLMI - Academic preparation (stat)	ST111 - Probability (Part A) (random variables, random processes, expectation, mean and variance, independence and conditional probability, law of large numbers)
MLMI - Academic preparation (inference)	ST111 - Probability (Part A) (maximum likelihood and Bayesian estimation) CS342 - Machine Learning (maximum likelihood and Bayesian estimation, regression, classification, clustering) CS910 - Foundations of Data Analytics (maximum likelihood and Bayesian estimation; regression; classification; clustering)
MLMI - Academic preparation (info theory)	none
MLMI - Academic preparation (comp)	CS118 - Programming or Computer Scientists (programming languages) CS126 - Design of Information Structures (data structures) CS136 - Discrete Mathematics and its Applications 1 (complexity) CS260 - Algorithms (algorithms) CS301 - Complexity of Algorithms (complexity) CS341 - Advanced Topics in Algorithms (algorithms)
MLMI - Academic preparation (project title)	An Investigation into the Application of Machine Learning to Create an Interface Between a Personal Event Schedule and Incomplete Event Requests
Summary of Available Marks	Overall: 81.5% (Year 1: 77.1%, Year 2: 80.3%, Year 3: 82.3%)

Application Information

Academic Awards

Best Overall Graduating BSc Student in Discrete Mathematics for 2017	Departmental prize for achieving the highest overall mark of 81.5% in my graduating class	30 Jun 2017
Discrete Mathematics Third-Year Project Prize for 2017	Awarded the departmental prize for the best third year project amongst my class with a mark of 84%	30 Jun 2017
Best Overall 2nd Year Discrete Mathematics student for 2016	Departmental prize for achieving the highest overall mark of 80.3% in my second year.	30 Jun 2016

Employment History

Sep 2019 -	Commoditisation and Delivery Engineer	Credit Suisse (London, United Kingdom)
Sep 2017 - Sep 2019	Technical Analyst	Credit Suisse (London, United Kingdom)

Other Applications Made

No other applications entered

Personal Information

Identifying Information

Full name Yerbury-Hodgson, Mr Oliver
Previous name

Date of birth 05 Sep 1995
Legal gender Male

Contact

Email oyerburyhodgson@gmail.com
Skype address
Contact address 2A Greenwich South Street,
Greenwich, London, Greater London,
SE10 8TY, United Kingdom
Valid until 01 Nov 2021

Phone 07910887587 (1st)
Home address Same as contact address
Valid until

Dependants

Partner WILL NOT bring partner

Child WILL NOT bring children

Disability

Disability No disability
Further information

Adjustment for Interview

Adjustment required No
Details

College Preferences

College St Edmund's College (1st)
Churchill College (2nd)

Current Membership

College Not College member

Funding Application

Cambridge Trust

Apply for Cambridge Trust Yes

Personal Statement

1437/3000 chars

While my education so far has largely been funded by a state school education and a small loan from Student Finance England, I have always put in all my efforts and excelled. I received an award for the highest final-year performance in my graduating class, giving me a great sense of achievement when completing my degree at Warwick University. The Discrete Mathematics (joint Computer Science and Mathematics) BSc begins with a grounding in core mathematical principles and the theory of computational complexity and optimisation. This culminates in the final year, where it was necessary to combine the applied teachings of modules with topics such as Machine Learning and computational parallelism with the foundations in statistics I'd learnt during the Mathematics portion of the degree. The dissertation I produced explored the use of Classifier Chains to populate missing attributes from personal calendar events. With positive results in performance against alternative Machine Learning models we'd studied, this too was recognised, receiving the highest graded dissertation in my cohort. This gave me confidence that my ability to communicate effectively and produce research of value had been proven at a bachelor's level. I believe that this master's will be yet another opportunity to not only educate myself, but also contribute to academic bodies like Cambridge University as well as the wider scientific community.

Research Councils

EPSRC, ESRC and STFC

- Engineering and Physical Sciences Research Council (EPSRC),
- Economic and Social Research Council (ESRC),
- Science and Technology Facilities Council (STFC)

You will automatically be considered by submitting this application form by your **funding** deadline.

MRC and NERC

- Medical Research Council (MRC),
- Natural Environment Research Council (NERC)

Refer to the appropriate website for more information.

Studentships

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Gates Cambridge Scholarships (Overseas)

Apply for Gates Cambridge No

Personal Statement

0/3000 chars

Your Funding

Funding Sources

No funding sources entered

Not circulated to department

The information in this section will not be circulated to departments.

Protected

Ethnicity White
Religion/Belief No religion

Sexual Orientation Heterosexual

Secondary Education

2nd Education Yes
School Type State school
Postcode at 17 SN47TA
Free Meals No

Care Leaver No
Caring Responsibilities No
Parents HE At least one had an undergraduate qualification

Visa Requirement

Not required

Passport

Not required

Study Visas

Visa not entered

Declaration

The information you have provided forms the legal basis of your application to the University of Cambridge. We reserve the right to refuse admission in the event of any misrepresentation by you. Submission of an application does not imply an offer of admission.

- The University of Cambridge, the Cambridge Colleges, the Gates Cambridge Trust and the Cambridge Commonwealth, European and International Trust (and their collaborators) will use your personal information for the purpose of processing your applications for admission and funding and deciding whether to offer you a place for the course you have applied for. For further information on the use of your personal information during the application process, please see 'How we use your personal information (for applicants)'.
- I certify that all the information given in this application is complete and accurate. I also understand that if I have given false or misleading information, the University of Cambridge will not admit me as a Postgraduate student and may take legal action against me.